Your broker shows what you own. We investigate whether you should still own it.

We are building **an AI investment team for everyday investors**, delivered through a beautiful financial terminal that works on their behalf.

The user explains their goals. The system researches and constructs a portfolio, remembers why each holding belongs, and monitors the evidence behind those decisions. When something material changes, specialist agents investigate, calculate the portfolio exposure and prepare adjustments within the user's limits.

The user can watch an investigation unfold or return later and replay it. Every decision connects to evidence, calculations, uncertainties and an approval record. The system can create targeted monitoring tools and improve its research procedures through tested memory.

The promise: start investing without becoming your own analyst, while being able to understand and verify what happens to your portfolio.

This is the consolidated specification. It incorporates the QA corrections. Where an implementation choice is still unresolved, it is explicitly identified rather than presented as settled.

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1. Who the product serves

The primary audience is **UK adults with money available to invest who have not started because they lack confidence, knowledge or time**. Early-stage investors who already hold a few investments are a natural secondary audience.

The FCA's 2024 survey found that 35% of UK adults held investment products under its measure excluding people whose only investments were property or other real assets. This establishes context for the participation gap; it does not mean everyone outside that group should invest immediately. www.fca.org.uk

The user's problem is:

> "I want to invest, but I do not know what to choose, how to judge the research, or when something has changed enough that I should act."

The product handles the complete lifecycle:

Understand the person → construct a portfolio → monitor its investment cases → investigate changes → maintain or adjust → review and improve.

It is approachable without removing depth. Someone can read the outcome or explore the agents' work and gradually learn through actual decisions. Education is a useful consequence, rather than a separate product.

The current scope is **ongoing, goal-constrained portfolio management**. Intraday strategy generation, correlation-only trading rules and reinforcement learning are outside this version.

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2. One functioning system, with an integrated historical demo

The product must operate against current markets. Historical demo mode is an environment inside that same application.

Behaviour	Historical demo	Current-market operation
Clock	Selected date; can pause or accelerate	Actual current time
Information	Available by the selected historical date	Latest available connected-source information
Agents	Real agents performing the work	The same agents
Monitoring	Scheduled against the simulated clock	Scheduled against the real clock
Calculations	Shared portfolio engine	Shared portfolio engine
Portfolio	Separate fictional historical portfolio	Persistent current-market paper portfolio
Memory	Isolated demo memory	Persistent current-market memory
Execution	Historical paper fills	Current-market paper execution
Label	“Historical demo · As of…”	“Current markets · Paper portfolio”

Turning off the demo changes the clock and data adapters. It must not replace prerecorded animations with an entirely different system.

Implementation requirements:

- Every time-sensitive service uses a shared clock.
- Data requests accept an `as_of` time.
- Records distinguish when an event occurred from when information became available.
- Historical mode blocks unrestricted current web retrieval.
- Historical holdings, dependencies and memories are versioned.
- Demo outcomes never enter current-market memory.
- Switching modes opens the corresponding portfolio and history.
- Every source shows freshness, delay and coverage.

Two different controls are needed:

- **Replay investigation:** revisit recorded events without rerunning the agents.
- **Run historical simulation:** execute the system through a historical period.

Current-market mode should genuinely research and monitor today's information. Real-money execution is a separate integration and deployment stage. The hackathon explicitly requires paper trading and prohibits real financial records, account details and logins.

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3. The first-time user flow

The user begins with a goal, not a ticker.

Onboarding collects:

- Goal and target date.
- Starting amount and expected contributions.
- Potential need to withdraw early.
- Investment experience.
- Ability to absorb losses.
- Willingness to tolerate losses.
- Asset or sector exclusions.
- Notification preferences.

For the hackathon, these are fictional profiles and amounts.

Use concrete questions:

> "If this portfolio fell from £5,000 to £4,000, would that prevent you from meeting your goal?"

The system distinguishes psychological tolerance from financial capacity.

It converts the answers into an editable mandate:

Mandate field	What it controls
Goal and horizon	Supported allocation policy
Access-to-cash needs	Liquidity and cash requirements
Risk capacity and tolerance	Allowed exposure ranges
Asset restrictions	Instrument eligibility
Concentration limits	Maximum direct and underlying exposures
Contributions	Purchases and rebalancing opportunities

Approval rules	Preparation versus execution authority
Review triggers	Circumstances requiring reassessment

The initial mapping must be a documented policy. Argon can explain it and identify contradictions; it should not invent a precise risk score.

If answers conflict, ask a relevant follow-up. If no supported policy fits, explain that rather than forcing a portfolio.

The user reviews the mandate and selects **Build my portfolio**.

Unresolved: the precise questionnaire-to-allocation mapping. The prototype needs an explicit, labelled policy; it must not be presented as a validated suitability assessment.

---

4. Where actual funds and stocks come from

The system selects from a **verified instrument catalogue**.

For the weekend, a proposed scope is approximately 20–40 instruments across broad equities, bonds and cash-like exposure, with a small separately constrained stock universe if included. This is an implementation limit, not the permanent product boundary.

Each record contains:

- ISIN and exact share class.
- Exchange-specific ticker.
- Issuer identity.
- Quote currency and fund base currency.
- Accumulating or distributing status.
- Asset class, objective and benchmark.
- Fees.
- Dated underlying holdings.
- Supporting documents and source references.
- Historical-data coverage.
- Eligibility and verification status.

Sources include:

Source	Purpose
Issuer product pages and documents	Objective, fees, share class, restrictions and disclosures
Issuer holdings files	Underlying company, sector and geographic exposures

Public exchange/broker listings	Listing identity and availability
Market-data provider	Prices, distributions, corporate actions and FX
Official announcements	Material changes to an instrument

For example, Vanguard's FTSE All-World accumulating ETF page supplies documents, portfolio information and purchase details. It illustrates a source-backed catalogue entry, not a universal recommendation. www.vanguard.co.uk

Agents extract fields. Code validates identifiers, units, dates and consistency. Unresolved identity conflicts or essential missing information block eligibility.

Ticker alone is insufficient. Different listings can represent the same share class, while similarly named funds can have different exposures. A GBP listing does not automatically hedge underlying currency risk.

If research discovers a candidate outside the catalogue, it enters **candidate review**. It becomes selectable only after the same identity, document, exposure and data-coverage checks. An agent cannot bypass the catalogue by inserting a new ticker directly into an order.

---

5. How the first portfolio is selected and calculated

Three jobs remain separate.

First: choose the intended exposure policy.

The mandate determines allowable equity, bond and cash ranges, plus concentration and liquidity requirements.

Second: compare eligible instruments.

Apply hard eligibility filters before comparison. Then assess:

- Whether the instrument provides the intended exposure.
- Overlap with other holdings.
- Fees and implementation characteristics.
- Exact share class and availability.
- Quality and freshness of supporting data.
- Restrictions and material risks.

The UI should answer:

> "Why this fund rather than the other eligible alternatives?"

Avoid a mysterious universal "AI quality score."

Third: calculate weights.

The proposed MVP optimiser balances:

- Distance from intended asset exposures.
- Estimated portfolio risk.
- Ongoing costs.
- Transaction costs.
- Unnecessary turnover.

Constraints include:

- Weights and cash sum to 100%.
- No shorting or leverage.
- Asset-class ranges.
- Holding and underlying concentration limits.
- Available cash.
- Supported order sizes and instruments.

CVXPY is suitable for constrained optimisation. Ledoit–Wolf covariance estimation is a reasonable initial numerical component for estimating risk from historical returns. www.cvxgrp.org

Independent code recomputes weights, cash, fees, currency conversions, overlap and constraint compliance. Checks run again after allocations become paper orders.

If no feasible portfolio exists, explain the conflict. Do not silently weaken the mandate.

This remains active management: research can change eligible instruments or proposed target ranges. However, the optimiser does not manufacture an investment advantage, and the model must not supply invented return forecasts as facts.

Unresolved: exact comparison weights, optimiser coefficients and policy ranges. These must be selected before final evaluation and kept fixed during it.

---

6. The initial research experience

The lead agent opens the main workspace:

> **Building your portfolio**
> Selecting eligible investments for your goal, comparing alternatives and verifying the allocation.

As tasks are delegated, the lead moves into the left half of the screen. Specialists appear on the right.

They work on concrete questions:

- Which instruments meet the mandate?
- Which candidates duplicate exposure?
- What do their documents establish?
- How has this allocation behaved historically?
- Which claims or assumptions remain unsupported?

The result is one recommended portfolio and, where useful, up to two alternatives with meaningful trade-offs.

The proposal shows:

- Allocation and each holding's purpose.
- Candidate comparisons.
- Costs and exposures.
- Historical behaviour, explicitly labelled.
- Material uncertainties.
- Passed and failed checks.

Actions:

Explore research · Compare alternatives · Change preferences · Approve paper portfolio

Approval saves the holdings and their **investment cases**: why each belongs, supporting evidence, material assumptions and conditions that could justify reconsideration.

---

7. How the system discovers portfolio dependencies

This happens during initial research and continues afterwards.

Step	Operation	Output
Resolve holdings	Identify exact funds and companies; load dated fund constituents	Portfolio → fund → underlying company links
Gather evidence	Retrieve filings, risk disclosures, announcements and relevant operational material	Relationship-bearing source documents
Extract candidates	Identify organisations, products, facilities and locations	Possible dependency records
Resolve entities	Check aliases, parents, subsidiaries and similar names	Correct entity identities
Verify relationships	Check whether a source actually supports the proposed connection	Documented, reported, inferred or unresolved relationship

Prioritise	Assess relevance, ownership exposure and evidence quality	Relationships worth further research
Create watches	Configure an existing monitor or request a new tool	Executable monitoring programme

A source might establish:

> Company B purchases a particular component from Supplier A.

That supports a relationship. It does not establish the effect of a disruption.

Each graph edge stores:

- Relationship type and direction.
- Exact supporting passage.
- Source and publication date.
- Date first known to the system.
- Applicable period.
- Verification status.
- Limitations and contrary evidence.

The graph is an evidence-backed research record, not established causal truth merely because an agent drew a line.

Undisclosed or unresolved dependencies remain unknown.

---

8. How relevant events are found

There are two routes.

Known-relationship monitoring

1. Research establishes a supplier relationship.
2. A saved watch detects a new announcement.
3. Entity matching connects it to the relationship.
4. A specialist investigates whether it changes the investment case.

New-relationship discovery

1. Broader ingestion finds an event involving a company, product, location or regulation.
2. Retrieval searches portfolio-company documents for potentially relevant exposure.
3. The lead asks:
   “Do any holdings depend on this facility, component or location?”
4. Specialists investigate candidate connections.
5. Supported relationships enter the graph.
6. New targeted monitors are created when useful.

The second route is essential. Without it, the system only detects risks it already anticipated.

Broader reviews and samples of discarded alerts also help measure what the filtering layer misses. No architecture should claim complete awareness of every possible dependency.

---

9. Exactly how monitoring is created

There are three monitor types:

Type	Creation	Execution
Standard checks	Developers implement tested functions	Normal scheduled code
Portfolio-specific checks	Argon configures existing templates	Standard monitor runner
New capabilities	A coding agent implements a missing adapter or function	Restricted, versioned execution

Standard checks include allocation drift, new filings, stale data and numerical limit breaches.

The creation pipeline is:

Investment assumption → observable evidence → monitor specification → template or generated tool → independent tests → activation → scheduled observations → investigation.

A monitor specification contains:

- The question being watched.
- Related investment case.
- Sources and entities.
- Extraction method.
- Trigger condition.
- Schedule.
- Review/expiry date.
- Resource budget.
- Escalation task.

For example:

```
None
{
  "monitor_id": "supplier_status_01",
  "investment_case_id": "case_17",
  "question": "Has the supplier's operating status changed?",
```

```
  "source_ids": ["supplier_announcements"],
  "runner": "announcement_change_detector",
  "entity_ids": ["supplier_a", "held_company_b"],
  "schedule": "every_6_hours",
  "event_types": [
    "production_suspension",
    "capacity_reduction",
    "reopening"
  ],
  "on_trigger": "launch_dependency_investigation",
  "review_after_days": 30
}
```

The schedule is illustrative.

A cheap runner checks for new material. A smaller model may assess relevance. Argon handles deeper interpretation.

For generated tools:

1. The lead specifies the evidence need.
2. A tool builder produces code, a manifest and sample outputs.
3. A restricted runner executes it.
4. Independent checks verify source identity, dates, units and extracted meaning.
5. Known examples test detection behaviour.
6. The monitor initially runs in shadow mode, recording detections without triggering downstream investigations.
7. An activation policy determines whether it can begin escalating.
8. Parser failures, excessive alerts or expired relevance can disable it.

Self-authored tests passing is insufficient. Use independently specified expected outputs.

Generated tools cannot alter portfolio limits, trading permissions or the independent financial engine.

Unresolved: exact activation thresholds, shadow period and cost limits. These are policy settings to test, not financial truths.

---

10. What happens while the user is away

Monitors run against the selected clock. Shared ingestion avoids fetching the same announcement separately for every portfolio.

Research begins when:

- Evidence may invalidate an investment case.
- An agreed numerical limit is breached.
- A scheduled review is due.
- The user changes their goals.
- An uncertainty becomes answerable.

Each run records its trigger:

> “Started because monitor M17 found a new announcement relevant to holding H4.”

The home screen reports:

Monitoring normally · Investigating · Awaiting approval · Monitoring degraded

Frequent monitoring does not require frequent trading.

After several quiet days, a returning user sees:

- Last completed review.
- Active watches.
- Source coverage and outages.
- Unresolved questions.
- A concise explanation of why no adjustment was proposed.

Notifications are reserved for meaningful decisions or material coverage problems. Routine activity can appear in a digest.

---

11. From an event to portfolio materiality

Keep three quantities separate:

Quantity	Meaning
Ownership exposure	How much the portfolio owns the affected company
Dependency severity	How important the affected relationship is
Possible investment impact	What the event could mean for the investment case and returns

Example:

- Portfolio holds 30% in Fund C.
- Fund C holds 4% in Company B.
- Portfolio also holds 2% directly in Company B.

Total exposure is:

\[30\%\times4\%+2\%=3.2\% \]

That is **3.2% ownership exposure**, not a predicted 3.2% loss.

The investigation considers:

- Whether the relationship is current.
- Importance of the affected product/process.
- Duration.
- Inventory and substitutes.
- Mitigation and contrary evidence.
- Direct and fund-overlap exposure.
- Whether the issue was already part of the original assessment.

Numerical exposure is recomputable. Economic significance remains a source-backed judgement with uncertainty.

---

12. How research becomes a proposed adjustment

The lead retrieves the mandate, portfolio, original investment case, triggering evidence and applicable memories.

It delegates specific questions and may launch child agents where uncertainty remains.

Possible outcomes:

Maintain · Investigate further · Prepare adjustment

Research can propose changes to:

- Instrument eligibility.
- Target allocation ranges.
- Exposure caps within the user's overall mandate.
- Scenario assumptions.
- Monitoring intensity or review dates.

Every proposed change cites its evidence.

The optimiser calculates a feasible implementation. The decision view compares current holdings, proposed holdings and no change.

Before proposing a trade, the policy should consider:

- Whether evidence is sufficient.
- Whether the change is material.
- Costs and turnover.
- Whether contributions can resolve drift.

- Minimum trade size.
- Recent related proposals and cooldowns.
- Uncertainty that warrants waiting.

Unresolved: the exact decision thresholds, no-action bands, trade-size bounds and cooldowns. These must be explicitly configured and evaluated.

The user sees:

Approve paper adjustment · Keep current portfolio · Ask a question · Request alternative

Before execution, revalidate holdings, prices and mandate version. Prevent duplicate execution. Record no-change decisions as carefully as trades.

---

13. Memory that changes future behaviour

Store four distinct memory types:

Type	Contents
User mandate	Goals, preferences and limits
Investment cases	Holding rationale and invalidation conditions
Investigation episodes	Evidence, findings, decisions and outcomes
Research procedures	Tested rules that change later behaviour

A later review asks:

- Was the original decision defensible with available information?
- Was relevant, obtainable evidence missed?
- Was the interpretation flawed?
- Was the process reasonable but the outcome unlucky?
- Would a proposed correction help elsewhere?

A worked example:

1. An investigation missed an available supplier disclosure.
2. Review identifies that specific omission.
3. A candidate rule requires a supplier-capacity check before increasing certain exposed holdings.
4. The rule is tested on other relevant and irrelevant cases.
5. It is assessed for useful detections, false alarms, cost and regressions.
6. If accepted, it becomes an active trigger.
7. A later proposed increase launches the supplier specialist automatically.

The rule stores its origin, scope, exceptions, tests, version and review date. It can be retired or rolled back.

The claim is **tested procedural improvement**, not automatic improvement merely from storing more memories.

Unresolved: promotion criteria and evaluation case sets.

---

14. Specific non-Gemini models

Model	Input → output	Useful role	Boundary
GLiNER	Text and entity types → entity spans	Find company, product, facility and location names	Does not independently establish relationships or identity
BGE-M3	Questions/documents → retrieval representations	Find relevant evidence and memories despite different wording	Similarity is not truth
BGE reranker	Question and passages → relevance ordering	Remove noisy retrieval results	Relevance is not financial confidence
FinBERT	Financial text → sentiment scores	Optional routing feature and comparison baseline	Sentiment is not a trade signal

These roles follow the models' documented capabilities. huggingface.co

Suggested starting choice: BGE-M3, with GLiNER if entity extraction needs it. Add the reranker if retrieval is noisy. FinBERT is useful as a conventional sentiment baseline.

Time-series models such as Amazon's Chronos family could be evaluated for suitable operational-series forecasts, but should not be added merely to claim stock-price prediction. github.com

Argon remains the lead investigator. Flash handles bounded parallel tasks. Numerical engines perform calculations. Each component has a defined job.

---

15. Suggested APIs and sources

Need	Suggested source	Qualification
Prices and instrument discovery	Twelve Data	Verify exact listings, history, adjustments and plan access

News and optional ETF metadata	Alpha Vantage	Test coverage for selected UK instruments
Broad news discovery	GDELT	Verify original evidence and archive availability
US company disclosures	SEC EDGAR	Use filing-specific availability in historical tests
Economic data and vintages	FRED/ALFRED	Distinguish information available then from later revisions
Fund characteristics and holdings	Issuer documents/downloads	Exact share-class and dated exposure verification
Public listing checks	Trading 212 catalogue	Identity and availability checks without account access
Evaluation	AgentVerse finance kit	Supplied cases, price history and hidden tests where applicable

The providers document these capabilities; that does not guarantee every selected instrument is supported on an available plan. twelvedata.com

Start with issuer documents, one price provider, one news source, SEC where relevant and the event's test data.

Test a representative equity ETF, bond ETF and stock before committing. Check identity, currencies, distributions, freshness, historical coverage and licence.

An internal paper ledger is sufficient. Broker integration is not required for current-market operation.

Unresolved: exact instrument universe and provider entitlements.

---

16. The beautiful visual workspace

The design reference is a financial terminal operated by agents. It contains persistent surfaces for portfolio data, prices, news, filings, dependencies, monitors and history.

The interface should feel approachable through hierarchy, not through removing information.

State	Layout
Start	Lead agent occupies the main canvas
Delegation	Lead takes the left half; specialists appear on the right

Work	Up to three readable specialist panels; additional branches in a compact rail
Inspection	Selected specialist expands, retaining its parent question
Child task	Enter the branch with clear ancestry
Review	Findings and contradictions return to the lead
Completion	Specialists collapse into accessible cards; lead expands with the result
Replay	Restore the workspace at a recorded event

Each panel shows:

Question → purpose → current action → evidence/output → finding and uncertainty.

Visual components:

Work	Display
News investigation	Headline, publisher, timestamp, permitted imagery and excerpt
Filing review	Document preview and highlighted passage
Claim checking	Supporting and contradicting evidence
Exposure calculation	Inputs, calculation and result
Dependency research	Labelled graph with source-backed edges
Tool creation	Purpose, output preview and expandable code
Tool testing	Test cases and results
Memory retrieval	Prior case/rule and why it applies
Scenario analysis	Assumptions and portfolio comparisons
Decision preparation	Options, checks and uncertainty

Clicking a finding should reveal **which source supported it, which calculation used it and which part of the proposal it affected**.

Do not use one generic “verified” badge. Separate:

- Source retrieved.
- Claim supported.
- Relationship current or uncertain.
- Calculation recomputed.

- Investment judgement uncertain.

Replay controls:

Previous · Next · Play · Jump to finding · Parent task · Open source · Compare versions

Changing an assumption creates a new branch. Completed branches remain accessible.

Use a restrained palette, strong typography, precise charts and deliberate motion. Source imagery must remain identifiable as source material. The interface represents observable work and concise explanations, not raw hidden model thoughts.

---

17. Structured events and screens

The backend emits typed events:

```
None
{
  "type": "source.inspect",
  "mode": "current_market",
  "run_id": "run_042",
  "agent_id": "supplier_analyst",
  "parent_agent_id": "lead",
  "sequence": 18,
  "title": "Checking alternative suppliers",
  "purpose": "Determine whether the disruption changes the
investment case",
  "investment_case_id": "case_17",
  "evidence_ids": ["filing_07", "announcement_12"],
  "status": "running"
}
```

The renderer selects the component. Actual tool execution produces lifecycle events; the model cannot fabricate a successful tool return.

Persist timestamps, ancestry, source versions, calculation inputs and results. Technical drawers can expose code, token usage and latency.

Required screens:

Screen	Main interactions
Workspace entry	Open current portfolio or historical demo

Goals and mandate	Edit answers, resolve conflicts, begin research
Agent workspace	Inspect, follow branches, pause/cancel, replay
Portfolio proposal	Compare, revise, approve
Portfolio home	Open holding, review investigation, change goal
Data/research explorer	Search, filter, inspect source, ask
Decision review	Approve, decline, question, request alternative
History	Replay runs, compare portfolio versions, review outcomes
Demo controls	Pause, advance, reset, exit historical mode

Reusable details cover holdings, fund comparisons, evidence, monitors, memory and settings.

The future scenario explorer lets users change assumptions and compare conditional portfolio paths. These are scenarios, not promised predictions.

---

18. Architecture and controls

A practical stack is React/TypeScript, Python/FastAPI, a database, background workers and isolated execution for generated code.

Required services:

- Shared clock and mode configuration.
- Data adapters and evidence storage.
- Instrument catalogue.
- Mandate and allocation engines.
- Agent orchestration.
- Dependency graph.
- Monitor compiler and scheduler.
- Restricted tool execution.
- Memory and procedure validation.
- Event store and streaming.
- Paper ledger and approval service.
- Evaluation harness.

Controls must cover:

- Unsupported instruments.
- Stale prices, holdings or mandates.

- Currency/unit mismatches.
- Concentration and cash limits.
- Costs and rounding.
- Duplicate orders.
- Missing data and outages.
- Generated-tool permissions.
- External content attempting to issue instructions.

The agents cannot alter the independent risk engine or grant themselves execution authority.

Argon's documented shared context capacity is 1,048,576 tokens. Use it for the relevant mandate, cases, filings, memories and specialist findings, while reserving capacity for outputs. docs.cloud.google.com

The compelling technical demonstration is a connection across time: an earlier filing establishes a dependency, a new announcement changes it, a tested memory rule identifies a missing check, and the lead launches that check before preparing a response.

---

19. Previous winners compared with this product

These comparisons use published descriptions, not independently verified investment performance.

Project	Published capability	Verified result	Comparison with yours
Counterfactual	Dependency graphs, hypothetical portfolio shocks, news impact and proposed paper hedges/trims	Winner, AI Broker Hackathon, Amsterdam, 18 June 2026. Five-person team; entrant count and prize amount not stated	Closest visual reference. Yours proposes a broader lifecycle: onboarding, construction, persistent cases, generated monitors and specialist replay
OrbitSatellite	Satellite imagery and news for commodity forecasts and supply-chain/procurement warnings	First place, London, 19 September 2026. **43 projects; €3,000 + US$2,000 DeepMind credits**	Strong reference for distinctive evidence. Yours focuses research around personal portfolio exposures
Autonomous Finance Portfolio Decision-Support Agent & Web Platform	Multi-agent finance platform using Google ADK for Java	**First place, in-person category**, AI DevCamp;	Demonstrates that this category can win. Your differentiation must be observable workflow

		organiser praises architecture	quality, not agent count
Lancaster Disclosure Degradation Detection System	Flags weakening or vague company and supplier disclosures	Winner, CFA AI Investment Challenge, 30 April 2026. **28 teams, 15 universities; £2,000**, plus finalist scholarships	Reference for a focused specialist capability. This was an investment competition, not a weekend hackathon

Sources establish the results and published descriptions. Unknown counts remain unknown; projects and teams are not participant counts. quantcompetitions.com

Counterfactual's opening is:

> "Bloomberg shows what happened. Counterfactual shows what happens next." quantcompetitions.com

Your opening uses that direct structure while describing your actual promise:

> **"Your broker shows what you own. We investigate whether you should still own it."**

Counterfactual has a cleaner single demonstration; Lancaster has a sharper narrow analytical claim. Your broader lifecycle becomes an advantage only if it works coherently. The visual investigation system can make that breadth understandable.

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20. Evaluation and the actual judging requirements

The Finance slides specify:

Criterion	Weight	Required evidence
Traceable reasoning	25%	Raw evidence to output, with maths separated
Unseen-data testing	25%	Suitable benchmark, costs and uncertainty
Controls and oversight	15%	Coded limits, logged actions, one control shown
Relevance and impact	20%	Specific user and consequential failure handled
Ambition and technical quality	15%	Adaptive work beyond a fixed workflow

The slides require paper trading, no real account details/logins, labelled synthetic inputs and no presentation of backtests as live performance.

Every team is assessed from its deck; only the top three pitch. Evaluation evidence belongs in the submission and a separate internal interface.

External comparisons:

Capability	Baseline	Measures
Portfolio management	Buy-and-hold, fixed rebalancing, suitable investable benchmark	Net returns, risk, drawdown, costs and turnover
Monitoring	Conventional price/entity/news alerts	Material detections, misses, false alerts and delay
Research	Generic assistant with comparable data and budget	Evidence accuracy, coverage and unsupported claims
Adaptive monitors	Fixed monitor library	Additional useful detections versus cost and false alerts
Traceability	Conventional reports/dashboard	Time and accuracy when verifying decisions

The planned historical experiment can span approximately five years **where the required data is available**.

It needs:

- Chronological development, validation and final-test periods.
- Historically eligible instruments and dated holdings.
- Evidence available at each simulated decision time.
- Frozen policy/model/monitor versions.
- Memory learned only from earlier information.
- Consistent contributions and cashflows across comparisons.
- Distributions, fees, FX, spreads and execution assumptions.
- Transparent missing-data handling.
- Uncertainty reporting and disclosure of tuning attempts.

Five years of prices alone is insufficient for five years of news-driven, dependency-aware evaluation.

Date filtering cannot erase subsequent events from model training knowledge. Historical replay and genuinely unseen testing must be reported separately. Use the event's hidden tests where applicable, plus current-market forward paper operation.

Memory/no-memory testing supports the procedural-improvement claim internally. The headline comparison remains the complete product versus existing approaches.

---

21. What is settled, what is proposed and what remains open

Decision	Status
Audience, product lifecycle and visual workspace	Agreed direction
Persistent cases, adaptive monitors and tested memory	Core requirements
Historical/current modes using the same system	Core requirement
Paper execution and human approval for the event	Required
Roughly 20–40 supported instruments	Proposed weekend scope
Constrained allocation rather than invented return forecasts	Proposed MVP approach
BGE retrieval and optional GLiNER/reranker	Suggested implementation choices
Exact onboarding allocation policies	Unresolved
Candidate ranking and optimiser coefficients	Unresolved
Trade thresholds, no-action bands and cooldowns	Unresolved
Monitor activation and resource limits	Unresolved
Memory promotion criteria	Unresolved
Exact instrument/provider coverage	Must be tested
Five-year evidence dataset and held-out cases	Must be assembled and checked

These open choices should be written into configuration and fixed before final testing. They should not be improvised by agents during evaluation.

---

22. The weekend build and demonstration

Preserve the complete lifecycle while narrowing breadth:

1. Open a clearly labelled historical scenario.
2. Establish a fictional mandate.
3. Show real instrument candidates and selection evidence.
4. Construct and independently verify the portfolio.
5. Approve paper holdings.
6. Show how a dependency was discovered and supported.
7. Advance to a held-out event.
8. Show the monitor that detected it and how it was created.
9. Split the workspace into specialist investigations.
10. Create and test one useful new monitoring capability.

11. Compare maintain versus adjust.
12. Demonstrate a coded limit blocking an invalid proposal.
13. Approve a valid paper adjustment.
14. Replay the decision back to its evidence.
15. Show a tested memory rule affecting a later investigation.
16. Exit historical mode and run the same system against current inputs.

Include a no-action case: an alarming headline that fails the materiality test should leave the portfolio unchanged.

The defining product moment: the user returns to a portfolio that has remained under active research, opens a proposed decision, and can visually follow what the team discovered, investigated, calculated, challenged and concluded.